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Subject Code: BBC202

Roll No:

BCA

(SEM. II) THEORY EXAMINATION 2025-26

DISCRETE MATHEMATICS

TIME: 3 HRS

M.MARKS: 70

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2x7 = 14

(a)	Let $A = \{1, 2, 3, 4\}$ and let $R = \{(1,2), (1,3), (4,2)\}$. Is R transitive?	CO-1	K3
(b)	Write the De-Morgan's law.	CO-1	K1
(c)	Write the Peano's axioms.	CO-2	K1
(d)	What do you mean by a Cyclic group?	CO-3	K1
(e)	Find the value of $(1 \wedge 0) \vee (0 \vee 1)'$.	CO-4	K3
(f)	What is the negation of the proposition " $3 + 5 = 7$ ".	CO-4	K2
(g)	What is the Isomorphic ordered set?	CO-5	K1

SECTION B

2. Attempt any three of the following:

7 x 3 = 21

(a)	Prove that $A \cap (B - C) = (A \cap B) - (A \cap C)$.	CO-1	K4
(b)	Solve $a_{n+2} - 4a_{n+1} + 4a_n = 2^n$	CO-2	K3
(c)	Let Z be the set of integers, show that the operation $*$ on Z , defined by $a * b = a + b + 1$ for all $a, b \in Z$ satisfies the closure property, associative law and the commutative law. Find the identity element. What is the inverse of an integer a ?	CO-3	K4
(d)	Construct the truth tables for the proposition: $(p \vee q) \wedge \sim (p \wedge q)$ and $(p \wedge \sim q) \vee (\sim p \wedge q)$. Are they equivalent?	CO-4	K4
(e)	Draw the Hasse diagram for the poset $(P(S), \subseteq)$ where $P(S)$ is the power set on $S = \{a, b, c\}$.	CO-5	K3

SECTION C

3. Attempt any one part of the following:

7 x 1 = 7

(a)	Consider the set Z . Let $a, b, c \in Z$. Define a relation $\leq$ on Z by requiring that $a \leq b$ iff a divides b .	CO-1	K2
(b)	Show that the $f: Q \rightarrow Q$ defined by $f(x) = 3x + 2$ is one-one onto, where Q is the set of rational numbers.	CO-1	K4

4. Attempt any one part of the following:

7 x 1 = 7

(a)	Prove that $\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$, for all natural numbers, $n \geq 2$.	CO-2	K4
(b)	Evaluate $1^2 + 2^2 + 3^2 + \dots + r^2$ using generating function.	CO-2	K5

5. Attempt any one part of the following:

7 x 1 = 7

(a)	Find the inverse of the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$.	CO-3	K3
(b)	In a group G , prove that $(ab)^2 = a^2b^2$ if $(ab)^{-1} = a^{-1}b^{-1}$ where $a, b \in G$.	CO-3	K4

6. Attempt any one part of the following:

7 x 1 = 7

(a)	Determine whether $(p \rightarrow q) \rightarrow [(p \rightarrow q) \rightarrow q]$ is a tautology.	CO-4	K4
(b)	Show that $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a contradiction.	CO-4	K4

7. Attempt any one part of the following:

7 x 1 = 7

(a)	Show that the lattice $(L^3, \leq_3)$ of 3 tuples of 0 and 1 is complemented.	CO-5	K4
(b)	If $L_1 = \{1, 2, 4\}$, $L_2 = \{1, 3, 9\}$ and $a \leq b$ if a divides b . Draw the Lattice $L_1 \times L_2$.	CO-5	K3